

Design and Analysis of an Additive Manufactured Supersonic Wind Tunnel

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For decades, supersonic wind tunnels have been designed and fabricated using common metallic materials for structural integrity. With advancements in additive manufacturing technology, high-performance thermoplastics are being developed as an alternative to steel that offer excellent strength and thermal stability. Utilizing the latest advancements in additive manufacturing, an additive manufactured supersonic wind tunnel with Mach 2+ capability was designed and fabricated using ULTEMTM 1010 resin for educational and research purposes at the University of Dayton Aerospace Laboratory. The supersonic wind tunnel features an open loop, blowdown design utilizing a high-pressure nitrogen cylinder with a settling chamber, modular C-D facility nozzle, modular 1" x 1" test section with optical access for flow visualization, and a step diffuser. High-fidelity CFD analysis was performed using CFD++ software on an AFRL supercomputer which verified the flowpath design estimated from method of characteristics and ideal operating conditions. The CFD analysis was performed with over 10.5 million cells and assumed compressible perfect gas while using a RANS 2-equation cubic k- ϵ turbulence model with viscous terms. Experimental tests were conducted using a Schlieren flow visualization system to validate the CFD analysis and characterize operating conditions for future educational and research activities.

I. Nomenclature

<i>AFRL</i>	= Air Force Research Laboratory
<i>CAD</i>	= Computer-Aided Design
<i>C-D</i>	= Convergent-Divergent
<i>CFD</i>	= Computational Fluid Dynamics
<i>CNC</i>	= Computer Numerical Control
<i>FDM</i>	= Fused Deposition Modeling
<i>LED</i>	= Light-Emitting Diode
<i>N-S</i>	= Navier-Stokes
P_0	= Total Pressure (psia)
<i>RANS</i>	= Reynolds-Averaged Navier-Stokes
<i>SWT</i>	= Supersonic Wind Tunnel
T_0	= Total Temperature (R)

II. Introduction

Additive manufacturing is a growing trend in the aerospace industry since it provides a low-cost alternative for smaller volumes and complex designs. It is a viable alternative to traditional CNC machining and other tooling processes that are often expensive and require a significant amount of time for manufacturing.¹ Additive manufacturing has enabled the ability to develop parts with complex geometries that are usually impossible to fabricate

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with traditional manufacturing processes. It offers design flexibility, weight savings, cost reductions, and improved lead times.¹ With advancements in additive manufacturing technology, various materials such as high-performance thermoplastics are being developed and becoming a viable alternative for traditional metallic or steel products. High-performance thermoplastics are developed with predictable mechanical, chemical, and thermal properties. One of the strongest FDM thermoplastics is ULTEM™ 1010 resin that possesses flame-retardant properties, superior tensile strength, high heat resistance and thermal stability, and has the lowest coefficient of thermal expansion of any FDM thermoplastic.^{2,3} It has previously been used for out-of-cabin aerospace components and under-the-hood automotive components.⁴

Several universities have SWT facilities to support educational and research activities for students and faculty. SWTs constructed using steel components require a significant monetary investment for design, manufacturing, and assembly in addition to operating and support costs. Additional hardware would likely be required for various test configurations during the lifetime of the tunnel facility. SWTs can be used to conduct fundamental research in supersonic fluid mechanics including supersonic flow around wedges, open cavities, and other geometries, along with studying shock wave and boundary layer interactions. Flow visualization techniques are often utilized to visualize the supersonic flow structure using a Schlieren system, surface oil flow visualization, or particle image velocimetry. This provides students an opportunity for hands-on activities and reinforces concepts being covered in undergraduate and graduate aerodynamics and compressible flow courses. However, in some cases universities may not be able to fund the initial investment, operational, and support costs of a traditional SWT. Utilizing advancements in additive manufacturing technology, an effort to design and demonstrate a low-cost, additive manufactured SWT for educational and research activities at the University of Dayton was carried out.

III. Design Process

Blowdown SWTs with large flowpaths require a substantial, high-pressure storage tank which increases total cost and space required for the facility. Typically, a blowdown SWT will consist of a high-pressure storage tank, settling chamber, C-D facility nozzle, test section, and a diffuser with atmosphere pressure at the tunnel exit.⁵ It was determined that the University of Dayton open loop, blowdown SWT would utilize a high-pressure nitrogen cylinder (Airgas Inc.) with a tank-mounted pressure regulator and commercial off-the-shelf nitrogen supply components shown in Figure 1. An alternative study was conducted to develop a pressure-vacuum blowdown tunnel with a high-pressure storage tank and high-flow-rate vacuum system downstream, but it drastically increased total cost and complexity of the tunnel design. Therefore, the design was reverted back to an open loop, blowdown SWT. The SWT has a 1" x 1" test section with optical access utilizing commercial off-the-shelf acrylic windows. The flowpath and corresponding SWT components were designed using Autodesk Inventor CAD software.

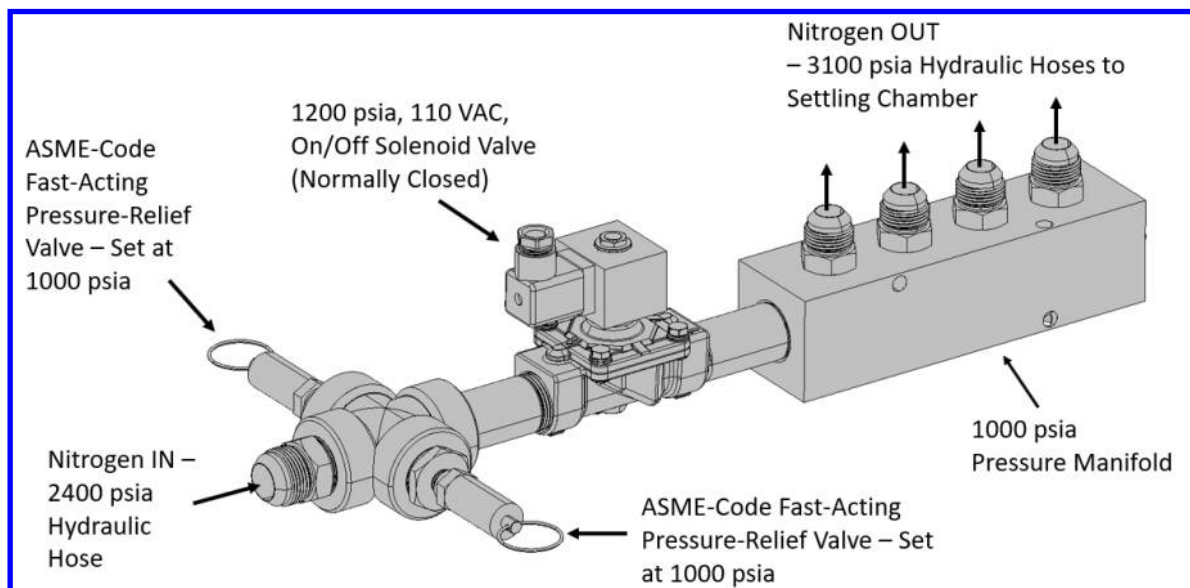


Figure 1. Nitrogen supply components for the University of Dayton SWT.

The C-D facility nozzle flowpath was designed using the NOZCS3 method of characteristics program from the University of Illinois to produce wave-free Mach 2 flow at the facility nozzle exit. The area ratio of the nozzle was determined by the area-Mach number relation assuming a specific heat ratio of 1.4 and a Mach design number of 2.05. The NOZCS3 program assumes inviscid flow for the method of characteristics calculations. However, after several iterations between nozzle design and CFD analysis, increasing the design Mach number at the nozzle exit by Mach 0.05 was sufficient for Mach 2 flow with viscous terms at the nozzle exit.

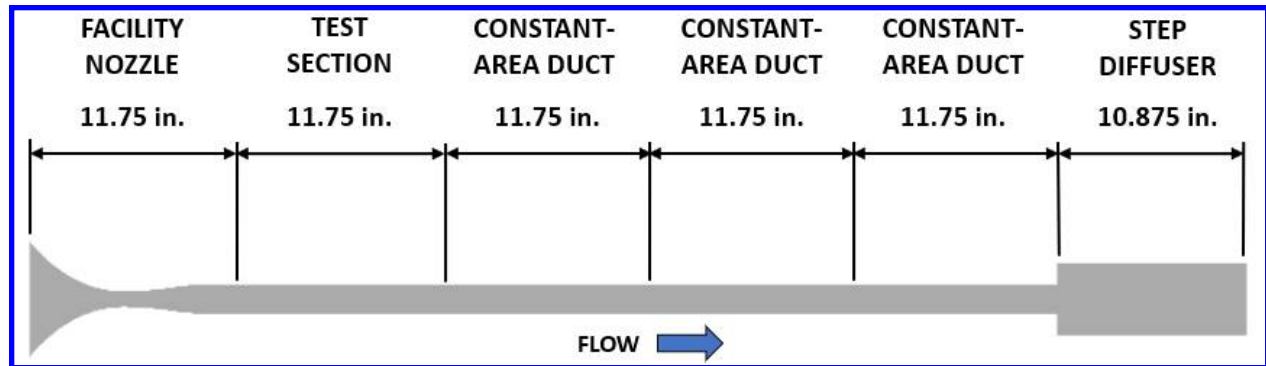


Figure 2. Illustration of the University of Dayton SWT flowpath.

A settling chamber was required upstream of the modular facility nozzle since it provides an area for the high-pressure nitrogen flow, injected at 0°, 90°, 180°, and 270° perpendicular to the axial flow direction, to settle and straighten out before entering the convergent section of the facility nozzle. The facility nozzle's convergent section increases the velocity of the incoming nitrogen flow until it is choked at the nozzle throat and enters the divergent section of the nozzle which increases the flow velocity to the desired Mach number of 2. The flow enters the modular test section which would be equipped with acrylic windows to allow for optical access and flow visualization. Downstream of the test section are constant-area ducts which allow the flow velocity to decrease to just below Mach 1 before reaching the step diffuser where the high subsonic flow expands and slows down to low subsonic velocities at the tunnel exit. Subsonic flow at the exit of an open-loop SWT is usually required for operational safety and reduces hazard levels for test personnel during testing.

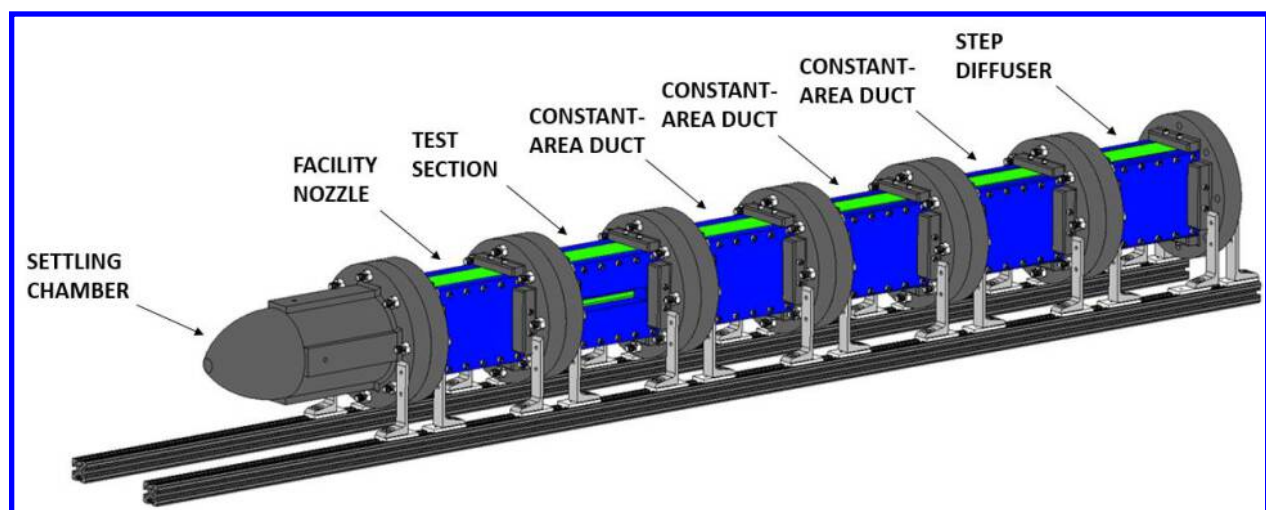


Figure 3. University of Dayton SWT components.

IV. CFD Analysis

Following flowpath design, high-fidelity CFD analysis was performed on the SWT flowpath to verify the design and ideal operating conditions. The flowpath geometry was imported into CFD++ on an AFRL supercomputer and a fully structured grid with over 10.5 million cells was used. The inflow boundary condition had a T_0 of 530 R with a P_0 range of 150, 225, and 300 psia for three separate cases. The outflow boundary condition consisted of a 14.7 psia back pressure to simulate realistic operating conditions since the flow exits into surrounding ambient air. The CFD model assumed compressible perfect gas (N-S/Euler) while using a RANS 2-equation cubic k- ϵ turbulence model with viscous terms. Convergence criteria were a three orders of magnitude decrease in the residuals and stable residual values for 1500 iterations. According to the CFD results discussed below, with the facility nozzle throat located at $x = 0$, the SWT started properly with P_0 values ranging from 150-300 psia with a constant T_0 of 530 R.

CFD results for the case with $P_0 = 300$ psia and $T_0 = 530$ R is discussed below. In Figure 4, the incoming subsonic flow is accelerated in the convergent section, becomes choked at the nozzle throat, and then accelerates to Mach 2 in the divergent section. Downstream of the nozzle exit, the flow gradually decelerates as it enters the test section and the constant-area ducts where the flow is expected to choke due to friction forces, similar to Fanno flow. Once choked, the high subsonic flow will enter the step diffuser where it expands and decelerates to low subsonic velocities. It is important to note that the skin friction coefficients for ULTEM™ 1010 components have not been characterized. The SWT flowpath with real internal viscous flow properties is expected to choke upstream of the step diffuser enabling subsonic flow at the tunnel exit. Since the SWT flowpath is entirely comprised of additive manufactured ULTEM™ 1010 components, the temperature distribution along the flowpath was critical for tunnel operability. In Figure 5, as the flow accelerates through the nozzle, the static temperature decreases to roughly 485 R at the nozzle exit. As the flow gradually decelerates downstream of the nozzle exit, the static temperature gradually increases to roughly 490 R prior to entering the step diffuser. The wall static temperature distribution along the flowpath was well below the maximum temperature of 860 R for ULTEM™ 1010.

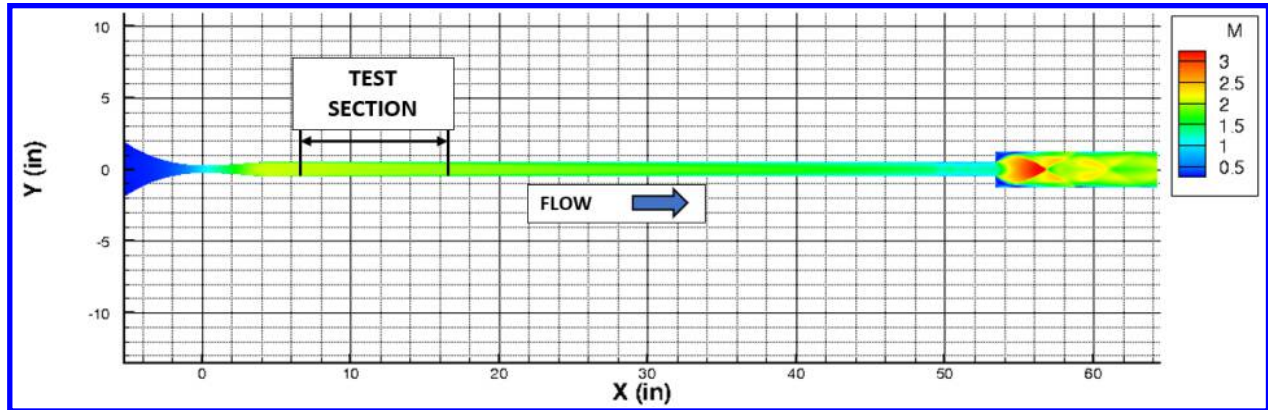


Figure 4. Mach number distribution along the SWT flowpath with $P_0 = 300$ psia and $T_0 = 530$ R.

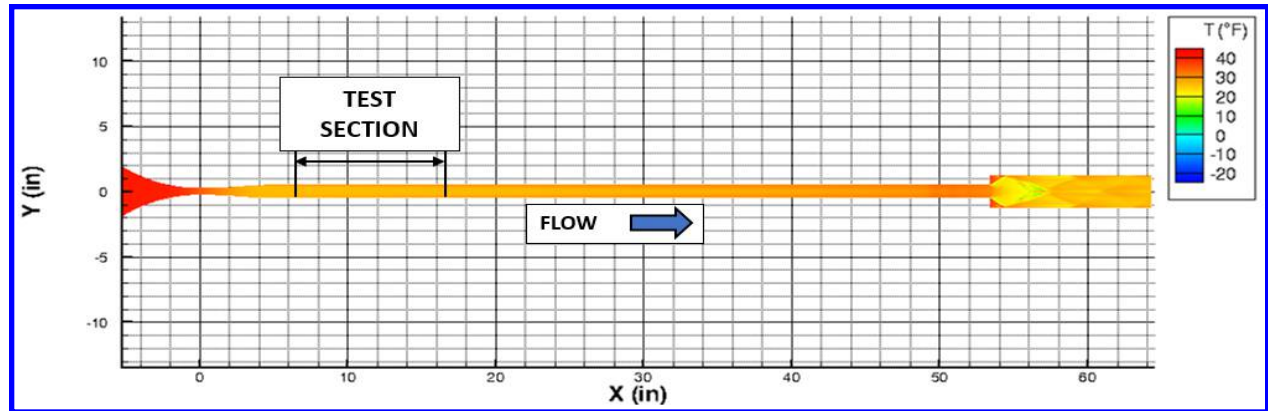


Figure 5. Static temperature distribution along the SWT flowpath with $P_0 = 300$ psia and $T_0 = 530$ R.

The centerline measurements for static pressure, Mach number, total pressure, and dynamic pressure along the SWT flowpath are shown in Figure 6. Similar to Figure 4, the Mach number accelerates in the nozzle to Mach 2 and then gradually decelerates downstream of the nozzle exit. As the flow accelerates through the nozzle, the total pressure remains relatively constant while the static pressure rapidly decreases. Meanwhile, the dynamic pressure rapidly increases, peaks near the choked flow at the nozzle throat, and then decreases towards the nozzle exit. Downstream of the nozzle exit, the total pressure remains relatively constant while the static and dynamic pressure gradually increase up to the step diffuser.

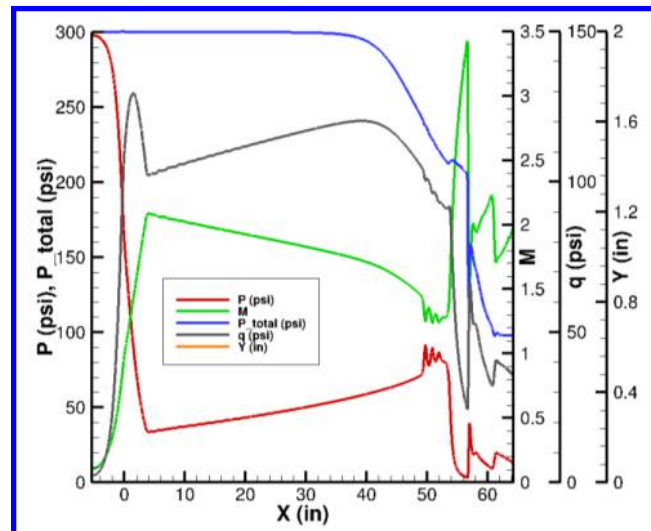


Figure 6. Centerline measurements along the SWT flowpath with $P_0 = 300$ psia and $T_0 = 530$ R.

V. Tunnel Characterization

The SWT was assembled and tested at the University of Dayton Aerospace Laboratory. The nitrogen supply system was initially tested separately to ensure all components were fastened and had no leaks prior to testing the SWT. To minimize leakage during operation, high-temperature graphite tape was installed around flowpath openings on mating surfaces. The nitrogen system was then installed, and safety checkout runs were conducted to ensure structural integrity of the SWT during operation. A Schlieren flow visualization system was then set up to verify the flowpath design. A 5-degree wedge was installed on the bottom of the test section to generate an oblique shock and verify supersonic flow in the test section. The Schlieren system included a concave mirror, LED flashlight with pinhole (light source), knife edge (light block), and a camera as shown in Figure 7. The Schlieren system works by using a mirror to focus a point source of light onto the knife edge which blocks incoming light. A camera is located behind the knife edge and focused on the test section in front of the mirror. When there is a change in the refractive index of the air in the test section, such as a shock wave generated by the wedge, the image of the light source will be slightly deflected. If the deflection is past the edge of the knife edge, the additional light captured by the camera will show up as streaks of light from the change in the index of refraction in the test section.

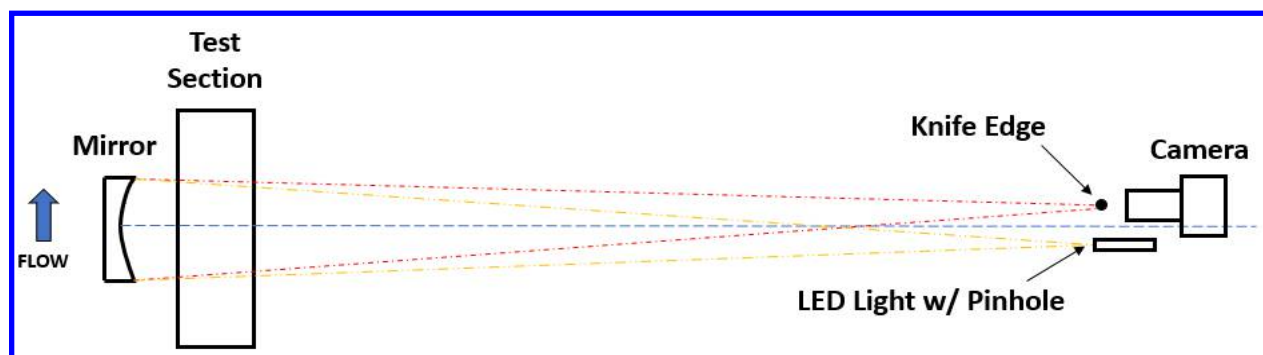


Figure 7. Visual representation of the Schlieren flow visualization system.

Based on the CFD analysis, the average Mach number along the centerline in the test section, which starts at $x = 6.422$ inches and ends at $x = 16.422$ inches excluding the flange sections in Figure 6, is approximately Mach 1.9. However, while taking real internal viscous flow properties into account the average Mach number is inferred to be around Mach 1.6-1.8. The 5-degree wedge was tested, and it generated an incident and reflected shock shown below with a supply pressure of 1000 psia at the exit of the tank-mounted pressure regulator. This experimentally demonstrated supersonic flow in the test section and further verified the flowpath design. Following several successful runs, the SWT was disassembled, inspected, and no structural issues were noted. Future efforts will be carried out to add instrumentation along the flowpath for improved characterization and health monitoring.

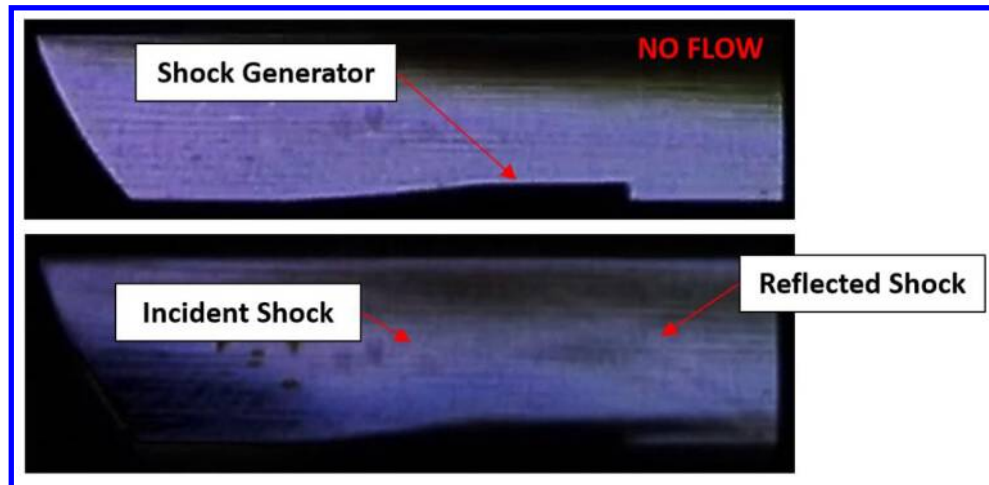


Figure 8. Schlieren image of the incident and reflected shock generated by the 5-degree wedge.

VI. Conclusion

A low-cost, additive manufactured SWT with Mach 2+ capability was successfully designed and demonstrated for educational and research purposes at the University of Dayton Aerospace Laboratory. The additive manufacturing material is a high-performance thermoplastic, ULTEM™ 1010, that is an alternative to steel with flame-retardant properties, superior tensile strength, high heat resistance and thermal stability, and the lowest coefficient of thermal expansion of any FDM thermoplastic. The University of Dayton SWT features modular components to enable various test configurations and a 1" x 1" test section with optical access for flow visualization. CFD analysis using CFD++ on an AFRL supercomputer verified the flowpath design and ideal operating conditions. Experimental tests were conducted using a Schlieren flow visualization system to validate the CFD analysis and characterize operating conditions for future educational and research activities.

VII. Acknowledgements

The authors gratefully acknowledge and appreciate Mr. Michael Mongin for his assistance in setting up the Schlieren flow visualization system at the University of Dayton Aerospace Laboratory and the computational resources provided by the AFRL Supercomputer Resource Center.

VIII. References

1. Hiemenz, J. (2016). Additive Manufacturing Trends in Aerospace. Retrieved from <https://www.sys-uk.com/wp-content/uploads/2016/10/White-Paper-Additive-Manufacturing-Trends-in-Aerospace-Leading-the-W....pdf>
2. Stratasy Ltd. (2019). ULTEM 1010 resin: The Strongest FDM 3D Printing Material. Retrieved from <https://www.stratasy.com/materials/search/ULTEM1010>
3. Stratasy. (2015). FDM Material Guide: Ultem 1010 Resin. Retrieved from http://producto3d.com/wp-content/uploads/2018/01/FDM_ULTEM1010_MG.pdf
4. Javelin Technologies Inc. (2019, May 27). ULTEM 1010 resin strong Stratasy FDM thermoplastic for 3D Printing. Retrieved from <https://www.javelin-tech.com/3d/stratasy-materials/ULTEM-1010/>.
5. Pope, A. and Goin K., *High-Speed Wind Tunnel Testing*, Wiley, New York, 1965

IX. Appendix

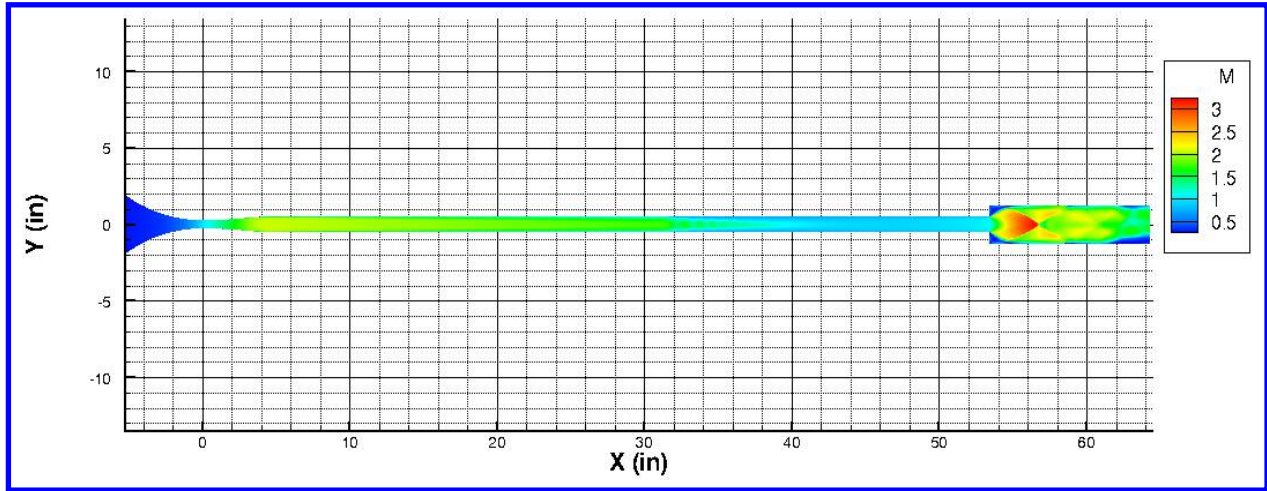


Figure A1. Mach number distribution along the SWT flowpath with $P_0 = 150$ psia and $T_0 = 530$ R.

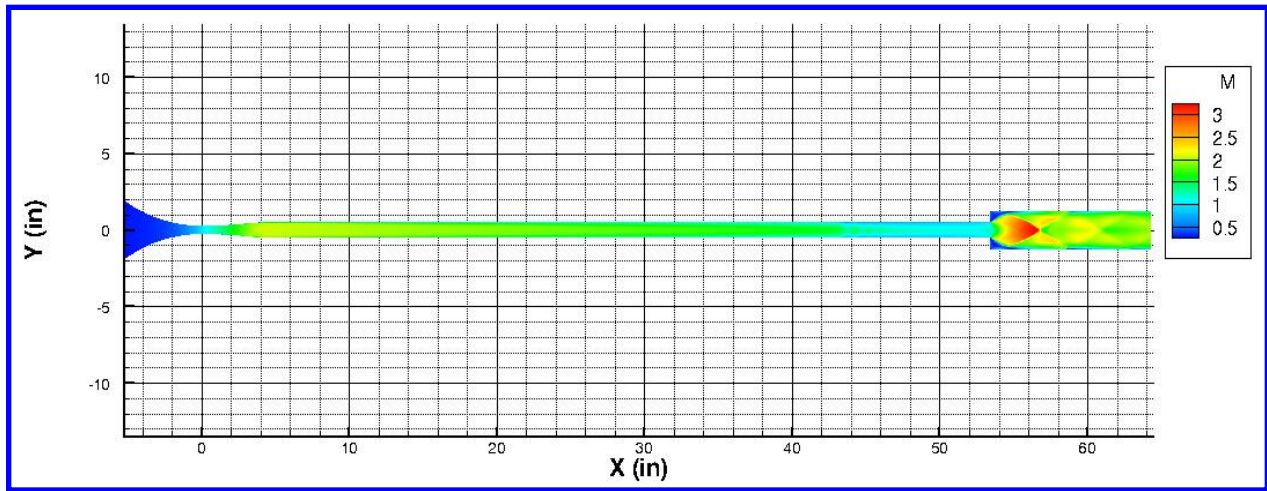


Figure A2. Mach number distribution along the SWT flowpath with $P_0 = 225$ psia and $T_0 = 530$ R.

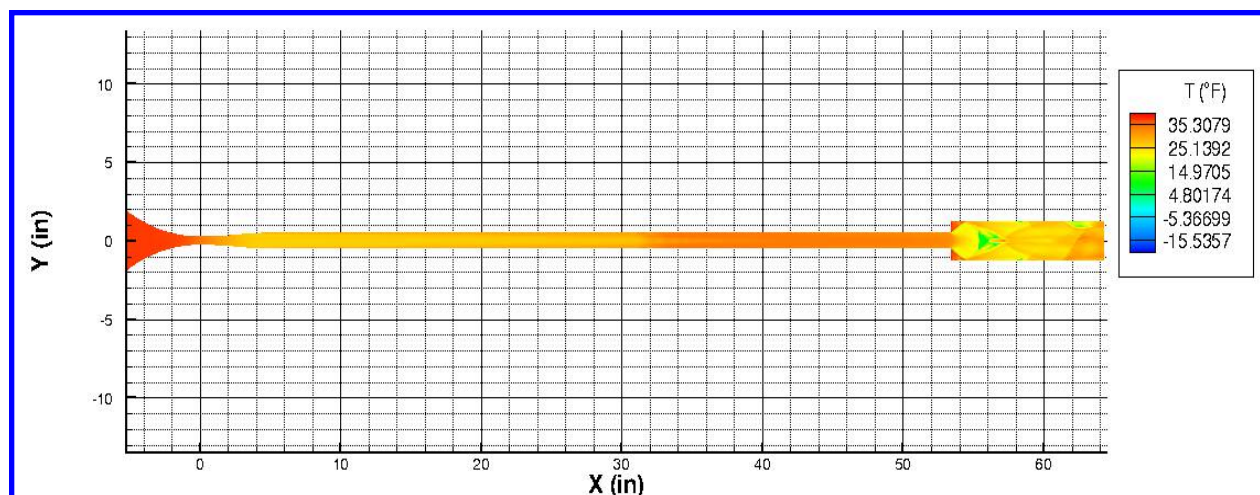


Figure A3. Static temperature distribution along the SWT flowpath with $P_0 = 150$ psia and $T_0 = 530$ R.

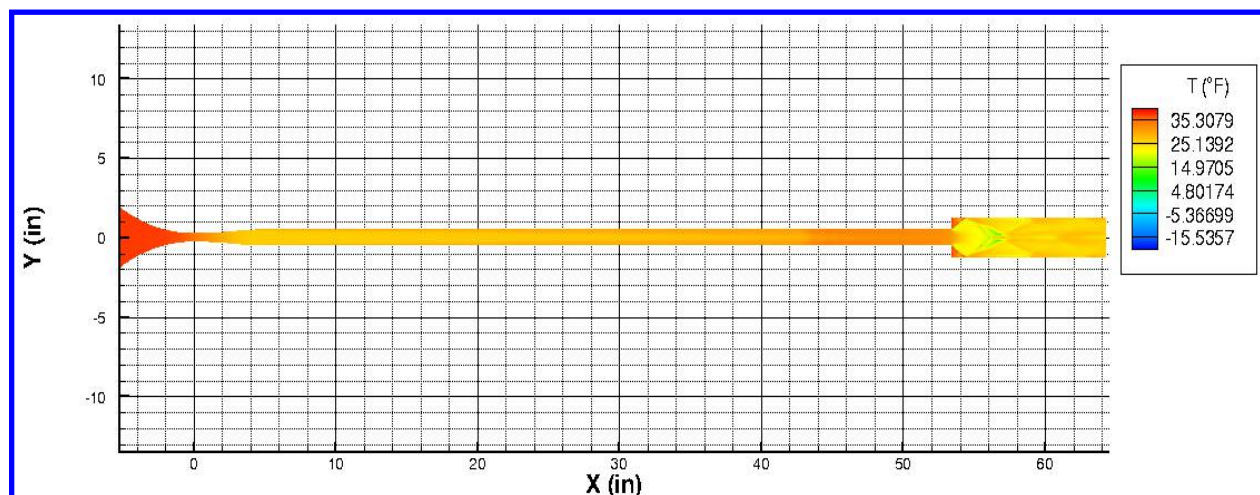


Figure A4. Static temperature distribution along the SWT flowpath with $P_0 = 225$ psia and $T_0 = 530$ R.

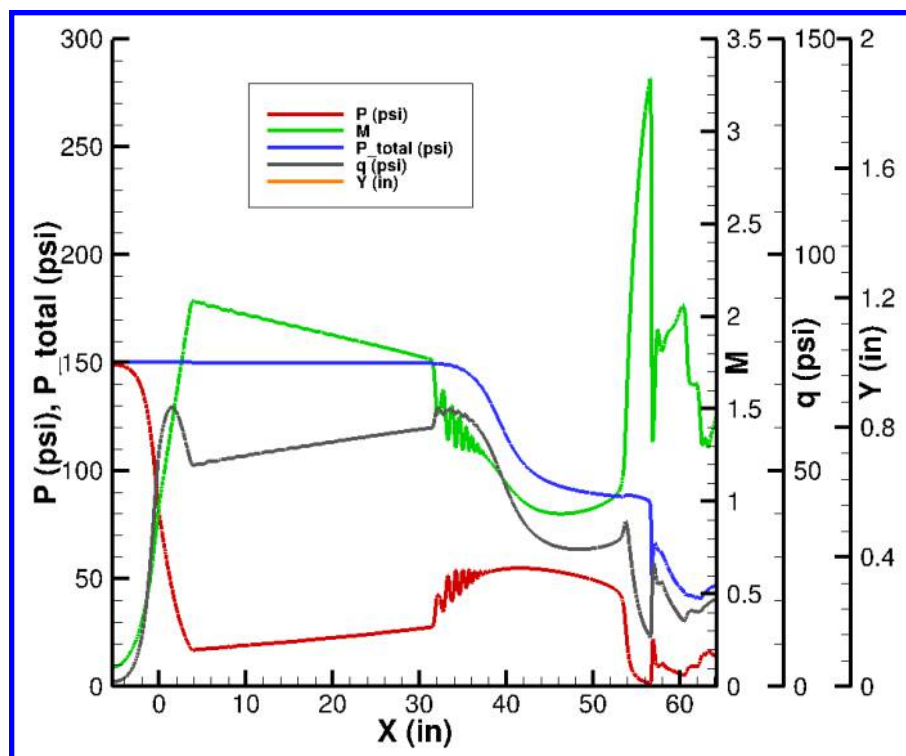


Figure A5. Centerline measurements along the SWT flowpath with $P_0 = 150$ psia and $T_0 = 530$ R.

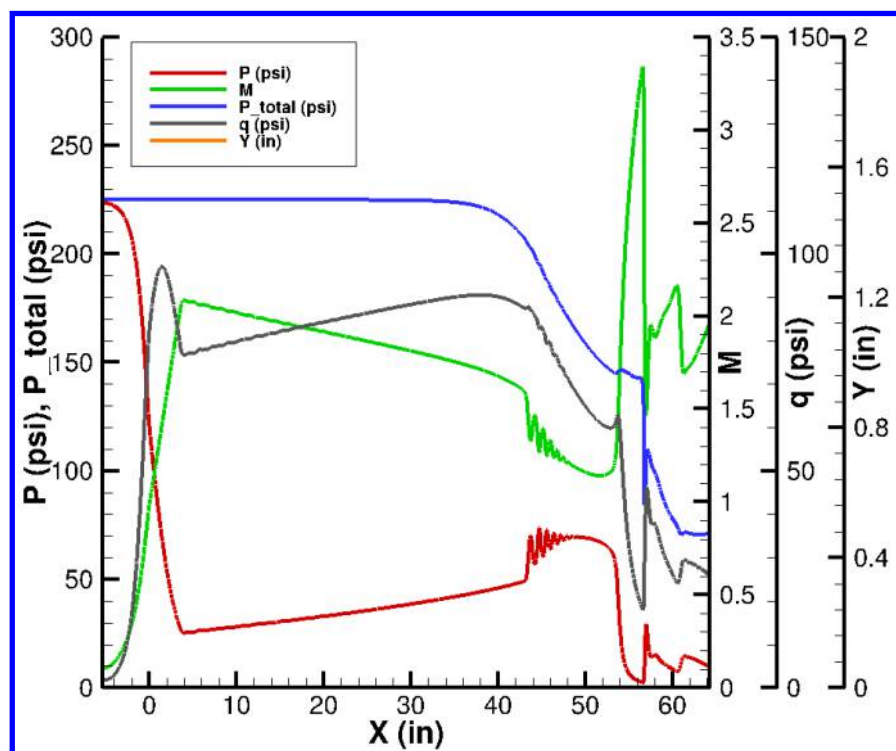


Figure A6. Centerline measurements along the SWT flowpath with $P_0 = 225$ psia and $T_0 = 530$ R.